

- Five percentile points
- The likelihood that the association is causal and whether the function includes that (*i.e.*, whether the function is conditional on the association being causal or unconditional)

The only information that we had about these distributions was the minimum, the maximum, and the five percentiles. The shape of the distribution was unknown. Therefore, we made an assumption that the cumulative distribution function (CDF) is piece-wise linear. In other words, we assumed that all values between the percentiles are equally likely. Following this assumption, we used linear interpolation between the percentile points to derive the CDF for each expert. We then made 15,000 draws from each CDF.

Table G-4 shows the inputs and the outputs of this process for each expert. The inputs are the minimum, the maximum, and the percentiles. The outputs are the percentiles that we calculated from the draws from the respective linearly interpolated CDFs.

Table G-4. Description of the Non-Parametric Expert Functions

Expert	Information	Min	P5	P10	P25	P50	P75	P95	Max
B1 4-10 $\mu\text{g}/\text{m}^3$	input	0.010	0.100		0.200	1.200	2.100	2.600	2.800
	output		0.099		0.203	1.213	2.092	2.599	
B2 >10- 30 $\mu\text{g}/\text{m}^3$	input	0.100	0.200		0.500	1.200	2.100	2.600	2.800
	output		0.198		0.501	1.191	2.096	2.597	
F1 4-7 $\mu\text{g}/\text{m}^3$	input	0.370	0.580		0.730	0.930	1.100	1.400	1.700
	output		0.581		0.732	0.928	1.097	1.407	
F2 >7-30 $\mu\text{g}/\text{m}^3$	input	0.290	0.770		0.960	1.100	1.400	1.600	1.800
	output		0.771		0.958	1.100	1.398	1.606	
H	input	0	0	0	0.400	0.700	1.300	2.000	3.000
	output		0	0	0.407	0.710	1.320	2.010	
L1 4-10 $\mu\text{g}/\text{m}^3$	input	0	0.200		0.570	1.000	1.400	1.600	2.700
	output		0.201		0.570	0.996	1.400	1.619	
L2 >10- 30 $\mu\text{g}/\text{m}^3$	input	0.020	0.200		0.570	1.000	1.400	1.600	2.700
	output		0.018		0.568	1.003	1.396	1.634	

Table G-5 below shows summary statistics for the draws from the non-parametric distributions that became BenMAP “custom” distribution tables. The section below on distributional details contains histograms for all the experts’ distributions.

Table G-5. Descriptive Statistics of the Random Draws from the Non-Parametric Expert Distributions

Expert	Mean	Standard Deviation	Min	P25	P50	P75	Max
B1 (cond)	0.01217	0.00897	0.00010	0.00200	0.01195	0.02090	0.02761
B1	0.01195	0.00901	0.00000	0.00195	0.01167	0.02075	0.02761
B2 (cond)	0.01290	0.00813	0.00100	0.00489	0.01187	0.02068	0.02761
B2	0.01262	0.00827	0.00000	0.00464	0.01159	0.02042	0.02761
F1	0.00937	0.00268	0.00370	0.00727	0.00924	0.01092	0.01686
F2	0.01144	0.00292	0.00290	0.00951	0.01091	0.01387	0.01784
H	0.00870	0.00662	0.00000	0.00406	0.00702	0.01302	0.02954
L1 (cond)	0.00985	0.00511	0.00001	0.00582	0.00999	0.01391	0.02662
L1	0.00739	0.00613	0.00000	0.00001	0.00727	0.01250	0.02659
L2 (cond)	0.00953	0.00544	0.00000	0.00567	0.00991	0.01389	0.02661
L2	0.00934	0.00549	0.00000	0.00531	0.00964	0.01371	0.02661

G.1.1.1.3 Using Expert Functions in BenMAP

When an expert has specified certain functional specifics with certain probabilities, the resulting “C-R function” becomes a set of possible functions, each with an associated probability. For example, expert K specified a piecewise log-linear function (i.e., two different log-linear functions on two different parts of the range of PM_{2.5}); this expert also specified a threshold within different ranges with different probabilities (and no threshold with a specified probability). BenMAP incorporates such a set of possible functions specified by an expert function by assigning appropriate weights to each specification. We illustrate this using expert K’s specification.

Expert K specified one log-linear function if the baseline PM_{2.5} value falls within the range from 4 µg/m³ to 16 µg/m³ and another log-linear function if the baseline value falls within the range from >16 µg/m³ to 30 µg/m³. BenMAP thus incorporates two sets of functions - one set for each of these two PM_{2.5} ranges - and selects from the set appropriate for a given PM_{2.5} baseline value. Expert K also specified a 64% probability that there is no causal relationship; an 18% probability that there is a causal relationship with no threshold, a 4% probability that there is a causal relationship with a threshold somewhere between 5 µg/m³ to 10 µg/m³, and a 14% probability that there is a causal relationship with a threshold somewhere between 0 µg/m³ to 5 µg/m³. Thus, the set of log-linear functions in BenMAP for expert K on the range from 4 µg/m³ to 16 µg/m³ contains a function with the:

- PM_{2.5} coefficient = 0 (no causality), which BenMAP selects with 65% probability;
- PM_{2.5} coefficient expert K specified for the log-linear function on that range and no threshold, which BenMAP selects with 18% probability;

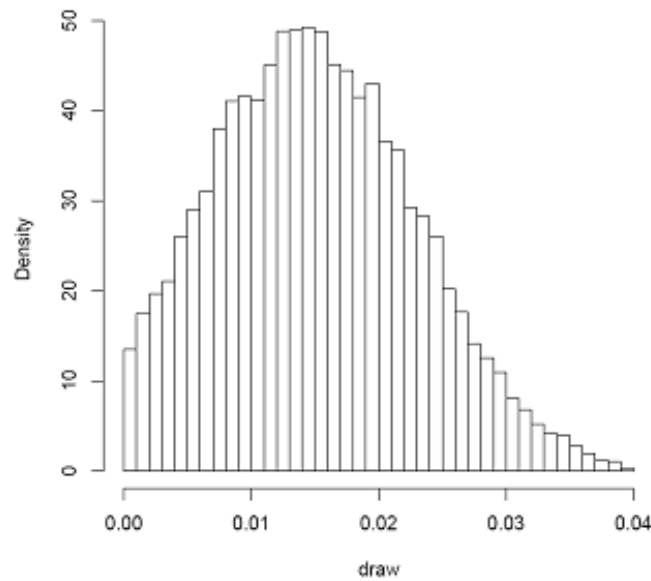
- PM_{2.5} coefficient expert K specified for the log-linear function on that range and a threshold (with uniform probability) between 0 µg/m³ to 5 µg/m³, which BenMAP selects with 14% probability; and
- PM_{2.5} coefficient expert K specified for the log-linear function on that range and a threshold (with uniform probability) between 5 µg/m³ to 10 µg/m³, which BenMAP selects with 4% probability.

If the PM_{2.5} baseline value is greater than 16 µg/m³, BenMAP goes through an analogous procedure to select a function from among the two functions in that set.

G.1.1.1.4 Distributional Details by Expert

Distributional details on each expert distribution are presented below. The derivation of the distributions is described above with additional details provided by Belova et al. (2007).

G.1.1.1.4.1 Expert A

Figure G-1. Histogram of the Random Draw from the Distribution of the PM_{2.5} Effect Specified by Expert A**Notes:**

Expert A specified a truncated Normal Distribution. We inferred the following values for the parameters of this distribution: mean=1.42 and standard deviation=0.89.

The experts specified distributions for the percent changes in the relative risk. The distribution of the corresponding PM_{2.5} effects was the following transformation of the percent change in relative risk Z - $\log(1+(Z/100))$.